

Indian Institute Of Information Technology Guwahati



Project Report 2

Group Number - 14

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Relational Design Report

Introduction:

This document presents the relational schema derived from the ER model of the Real Estate Management System. The schema has been refined to eliminate redundancy, enforce integrity, and ensure normalization up to higher normal forms.

Relations (Tables Definition)

(Primary Keys are underlined, Foreign Keys are specified clearly)

1. *User(**UID**, Name, Email, Phone, Password, ProfileURL, RegDate, Area, City, Pin)*
 - > Stores all platform users including buyers, sellers, and agents (agents are specialized users).
2. *Agent(**AID**, OfficeID, Rate, UserRating, License, Status)*
 - > FK → AID referencing User(UID)
 - > Stores additional details specific to agents.
3. *Office(**OfficeID**, Password, Email, Telephone, City)*
 - > Represents real estate offices managing agents and listings.
4. *Property(**PID**, OwnerID, OfficeID, HouseNo, Locality, Area, City, Pin, Size, Type, BHK, YearBuilt)*
 - > FK → OwnerID referencing User(UID), OfficeID
 - > Stores details of properties (houses/apartments)
5. *ListingToken(**TokenID**, PID, AID, ListingDate, ListType, Price, Description, Status)*
 - > FK → PID, AID
 - > Represents listings (rent/sell) created by agents for properties.
6. *Transaction(**TID**, AID, TokenID, BuyerID, Amount, Date, Type, Mode)*
 - > FK → AID, TokenID, BuyerID referencing User(UID)
 - > Stores completed deals (rent/sale transactions).
7. *Performance(**TrackID**, AID, Deals, TotalSales, Score, UserRating, DealsLeft)*
 - > FK → AID
 - > Tracks agent performance metrics.
8. *Notification(**CODE**, SID, RID, Message, Type, Date, Time, Status)*
 - > FK → SID, RID referencing User(UID)
 - > Stores notifications/messages between users.
9. *Images(**ImgID**, PID, URL, Date)*
 - > FK → PID
 - > Stores property images.

Normalization

Functional Dependencies (FDs)

Examples:

- $UID \rightarrow \text{Name, Email, Phone, Password, ProfileURL, RegDate, Area, City, Pin}$
- $AID \rightarrow \text{OfficeID, CommissionRate, LicenseNumber, Experience, Status}$
- $PID \rightarrow \text{OwnerID, OfficeID, HouseNo, Locality, Area, City, Pin, Size, Type, BHK, YearBuilt}$
- $\text{TrackID} \rightarrow \text{Deals, Score, TotalSales, UserRating, DealsLeft}$
- $\text{OfficeID} \rightarrow \text{Email, Telephone, City}$
- $\text{TokenID} \rightarrow \text{PID, AID, ListingDate, ListType, Price, Description, Status}$
- $TID \rightarrow \text{AID, TokenID, BuyerID, Amount, Date, Type, Mode}$

First Normal Form (1NF):

A relation is in 1NF if,

- All attributes contain atomic values
- No repeating groups exist
- Each record has a primary key

Evaluation: All relations (USER, AGENT, PROPERTY, LISTINGTOKEN, TRANSACTION, PERFORMANCE, NOTIFICATION, IMAGES, OFFICE) contain atomic attributes and defined primary keys. Therefore all tables satisfy 1NF.

Second Normal Form (2NF):

A relation is in 2NF if,

- It is already in 1NF
- No partial dependency.

All non-key attributes are fully dependent on the entire primary key

Evaluation: All tables use single attribute primary keys (UID, AID, PID, TokenID, TID, TrackID, Code, ImgID, OfficeID).

Since there are no composite keys, partial dependencies cannot occur.

Thus all relations satisfy 2NF.

Third Normal Form (3NF):

A relation is in 3NF if,

- It is in 2NF
- No transitive dependency exists
(non-key attribute determining another non-key attribute).

Evaluation: All the non-key attributes depend on the primary key (UID, AID, PID, TokenID, TID, TrackID, Code, ImgID, OfficeID) and not on any other non-key attribute. Thus it holds the 3NF.

Boyce-Codd Normal Form (BCNF)

- A relation is in BCNF if for every functional dependency $X \rightarrow Y$, X is a superkey of the relation.
- If LHS is not the superkey then it is not in BCNF

Evaluation:

Since all the functional dependency are of type: $PK \rightarrow$ all other attributes, where PK is the Primary Key, Hence the LHS will always be Super Key thus all relations satisfies BCNF.

Observations and Improvements

Possible improvements:

- Removed some derived attributes from the previous model
- Optional address normalization by introducing an ADDRESS relation.
- Optional geographic normalization by introducing a PINCODE relation if strict city-pin

Individual Contribution

1. Anmol Kumar (2401031)

Checked the data consistency and matched the constraint of the data, suggested possible edits

2. Anmol Kumar (2401032)

Reduction to relational database from ER modal by applying all the DBMS rules and maintaining all constraints, FD and Normalization and design backend.

3. Ansik Singh Tomar (2401037)

Prepared the spreadsheet containing the populated dataset while ensuring all specified constraints were satisfied, and drafted this report.

4. Anuved Pratap Singh (2401041)

Written the java code and mapped the Entities using JPA to form the database Schema for the mysql

Schema

Performance	
PK	<u>TrackID</u>
FK	AID Deals TotalSales Score UserRating DealsLeft

Notification	
PK	<u>CODE</u>
FK	SID
FK	RID Message Type Date Time Status (read/unread)

ListingToken	
PK	<u>TokenID</u>
FK	PID
FK	AID Listing Date ListType (Rent/Sell) Price Description Status (Open/Closed)

Agent	
PK - FK	<u>AID (=UID)</u>
FK	OfficeID Rate Status (Active/Inactive)

Property	
PK	<u>PID</u>
FK	OwnerID
FK	OfficeID HouseNo Locality Area City Pin Size Type (Flat/ Apartment) BHK Year Built

Images	
PK	<u>ImgID</u>
FK	PID URL Date

Office	
PK	<u>OfficeID</u>
	Password Email TelePhone City

Transaction	
PK	<u>TID</u>
FK	AID
FK	TokenID
FK	BuyerID Amount Date Type Mode

USER	
PK	<u>UID</u>
	Name Email Phone Password ProfileURL RegDate Area City Pin